

Prescribed Finite Étale Algebras as Full Fibers of Keller Maps with Symmetric Monodromy

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Abstract

Let K be a field of characteristic zero. We prove that every finite étale K -algebra of rank at least three occurs as the full fiber of an explicit polynomial Keller map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ with Jacobian one. The construction preserves the algebra itself, not only its rank or its geometric points. It gives an exact fixed-degree classification: for $N \geq 3$, the full fibers of geometric-degree- N Keller maps in $\mathbb{A}_K^3$ are precisely the rank- N finite étale K -schemes. Rank one is realized by the identity, while rank two is excluded by the Galois case of the Jacobian theorem. Thus the attainable nonzero ranks are exactly $1, 3, 4, 5, \dots$

The engine is a localized quadratic-gauge fiber theorem. For an arbitrary target with inverse polynomial E , it identifies the fiber as

$$\mathrm{Spec} \left((K[S]/(E))[1/\overline{E}] \right).$$

When E is squarefree of degree N , the localization disappears. To prescribe an algebra, start with a squarefree polynomial P of degree $N \geq 3$, choose a with $P'(a)P'''(a) \neq 0$, and insert $G(S) = P(a + S) - P(a)$ into the gauge. The resulting map has coordinate degree at most $6N + 2$; at a distinguished target its inverse polynomial is $P(a + S)$, and the literal scheme fiber is naturally isomorphic to $\mathrm{Spec} K[T]/(P)$. A direct function-field comparison proves that the actual geometric degree is N . The cubic seed $G(S) = S + S^3$ recovers the recently announced three-dimensional map coordinate for coordinate. Geometrically, this seed is an affine chart of the normalized marked-factor map $(L, Q) \mapsto LQ$ with $\mathrm{Res}(L, Q) = 1$: the three generic inverse branches mark the three possible linear factors of a cubic, while the nonvanishing resultant makes the reconstruction locally invertible. The quadratic gauge retains this marked-root mechanism while replacing the cubic inverse equation by the arbitrary polynomial G .

For every admissible seed, not merely for a generic one, the splitting field of the generic inverse polynomial over $K(\Pi, B, C)$ has geometric and arithmetic Galois group S_N . The proof produces a Morse vertical slice from a birational parametrization of the repeated-root curve. Thus every prescribed algebra occurs as a special fiber of a maximally symmetric generic cover. Over a Hilbertian field, each constructed map has infinitely many connected full fibers whose splitting fields have group S_N . The reduced nonproperness locus is exactly the zero set of the inverse discriminant. Direct charts give every fiber on $\Pi = 0$, while a uniform Newton calculation gives the exact global discriminant order $N^2 - 3N - 2$ for $N \geq 4$.

Connectedness, field decomposition, Galois action, real signature, local points, and scalar extension pass to the Keller fiber unchanged. Over $\mathbb{Q}$, the classical Berend–Bilu quintic gives an explicit rank-five fiber with points over $\mathbb{R}$ and every $\mathbb{Q}_p$, but no rational point; rank five is minimal.

The polynomial-presentation part is formalized in Lean: the automatic choice of a , the actual three-variable polynomial map, its Jacobian-one identity, the $6N + 2$ degree bound, the source-function-field comparison and geometric degree N , and the natural equivalence between the literal map fiber and $K[T]/(P)$, including finite étaleness and rank. Independent SymPy and

Singular calculations check the coordinate identities, Jacobian, degree profile, concrete quotient reconstruction, and nonproperness boundary from separate implementations. The nonproperness theorem itself remains an ordinary proof rather than a Lean theorem. For the explicit quintic, Lean also certifies the absence of rational points and the existence of points over $\mathbb{R}$ and every $\mathbb{Q}_p$; only the Chebotarev-based rank-minimality argument remains outside that Hasse certificate.

1 Introduction

A polynomial map with nonzero constant Jacobian is étale. Its fibers are therefore finite étale schemes, possibly empty. This raises a simple reverse question: given a finite étale algebra A/K , can we build a Keller map with $\text{Spec } A$ as one of its fibers?

There is one important word in that question. The fiber should be *full*: its length must equal the geometric degree of the map. Otherwise some generic inverse sheets may have escaped to infinity, and the special fiber records only part of the inverse problem. Our main result answers the prescribed-algebra question. Every finite étale K -algebra of rank at least three occurs as a full fiber of an explicit Keller map in three variables.

Together with the elementary fact that every Keller fiber is finite étale, this gives an exact fixed-degree classification. For $N \geq 3$, the full fibers of geometric-degree- N Keller maps in $\mathbb{A}_K^3$ are precisely the rank- N finite étale K -schemes. The degree-two case is empty by the Galois case of the Jacobian theorem, while the identity realizes rank one.

For rank $N \geq 3$, the construction starts from a squarefree presentation

$$A \simeq K[T]/(P).$$

Choose $a \in K$ with $P'(a)P'''(a) \neq 0$ and set

$$G(S) = P(a + S) - P(a).$$

The polynomial G is inserted into one explicit three-variable quadratic-gauge map. A fixed linear change in the target makes its Jacobian one, and its coordinate degrees are bounded by $6N + 2$.

The family begins with the announced counterexample itself. The seed

$$G(S) = S + S^3$$

gives Alpöge’s announced map coordinate for coordinate, with no conjugation or output rescaling. The higher-degree construction is therefore an extension of that map rather than a separate template with the same Jacobian.

The geometric origin of this seed is the multiplication map for binary forms

$$(L, Q) \longmapsto LQ, \quad \text{Sym}^1(K^2) \times \text{Sym}^2(K^2) \longrightarrow \text{Sym}^3(K^2).$$

The relative scaling $(L, Q) \mapsto (\lambda L, \lambda^{-1}Q)$ is removed by the normalization $\text{Res}(L, Q) = 1$. A generic cubic has three possible marked linear factors, so the normalized map has three generic inverse branches; the nonvanishing resultant separates the marked root from the residual quadratic and gives local invertibility. In a suitable affine three-space chart this is precisely the announced cubic map [12]. The quadratic gauge introduced below keeps the same marked-root reconstruction but allows the scalar inverse equation to be an arbitrary polynomial G .

The displayed coordinates are fairly large, but the inverse problem is deliberately small. It is organized around

$$S = \frac{x}{1 + xy}$$

and reduces to one polynomial equation

$$E_{\Pi,B,C}(S) = 0.$$

At a target with nonzero first coordinate, the localized quadratic-gauge theorem identifies the entire fiber as

$$\mathrm{Spec} \left((K[S]/(E))[1/\overline{E'}] \right).$$

This is the right answer even when E has repeated factors: the source map is étale, so only roots at which E' is invertible can occur. At the prescribed target, $E = P(a + S)$ is squarefree, hence E' is already a unit modulo E and the localization disappears.

The same coordinates identify the actual function-field extension. On the dense chart $1 + xy \neq 0$, the source field is

$$K(x, y, z) = K(\Pi, B, S),$$

and S has minimal polynomial $E_{\Pi,B,C}$ over $K(\Pi, B, C)$. Since that polynomial is irreducible of degree N , the displayed map has geometric degree N . This comparison is what turns the literal rank- N fiber into a full fiber.

The generic inverse problem is also as large as possible in its Galois theory. For every admissible degree- N seed, not just a generic choice of its coefficients, the geometric and arithmetic Galois groups of $E_{\Pi,B,C}$ are both S_N . The proof fixes Π , parametrizes the repeated-root curve, and shows that all but finitely many vertical slices are Morse. The classical Morse-polynomial theorem gives S_N on one slice, and a specialization lemma embeds that group into the generic geometric group. Over a Hilbertian field, the same fixed map consequently has infinitely many connected full fibers with splitting-field group S_N .

The same inverse equation detects precisely where affine sheets are lost. Over an algebraic closure, the reduced nonproperness locus is

$$V_{\mathrm{red}}(\mathrm{Disc}_S E_{\Pi,B,C}).$$

This statement is set-theoretic, or equivalently an equality of reduced closed subschemes; it does not identify the generally nonreduced discriminant scheme with a scheme structure on the nonproperness locus. The slice $\Pi = 0$ is treated separately by direct source charts and a Newton polygon calculation.

The reconstruction is stronger than a count of geometric points. Its formulas are natural in the test algebra, so they identify the whole fiber functor, including nilpotents and every scalar extension. After translating back from S to T , the literal fiber is the original algebra $K[T]/(P)$. Nothing arithmetic has been replaced by a numerical proxy.

The proof has six parts. First, a coefficientwise calculation builds the all-degree map and proves its constant Jacobian and degree bound. Second, the root-to-source formulas give the localized fiber theorem and its squarefree corollary. Third, irreducibility and the explicit source-field comparison give the actual generic degree. Fourth, the repeated-root parametrization, the Morse-polynomial theorem, and specialization give full symmetric monodromy. Fifth, direct boundary charts, the characteristic-zero fiber-cardinality criterion, and a uniform discriminant calculation identify the exact reduced nonproperness locus. Finally, translation prescribes the polynomial quotient, while monogenicity passes from polynomial quotients to arbitrary finite étale algebras. The Campbell–Razar–Wright theorem separately excludes geometric degree two.

The arithmetic applications are then literal transfers. An irreducible polynomial gives a connected fiber; a product gives the same field decomposition; local points, real signature, and Galois action are inherited from the original algebra. In particular, the classical Berend–Bilu quintic produces a rank-five Hasse-principle failure, and a short Chebotarev argument shows that rank five is minimal over $\mathbb{Q}$.

The constructive core is formalized in Lean. Starting from squarefree P of degree at least three, Lean chooses an admissible translation, constructs the actual three-variable polynomial map and target, proves the Jacobian-one identity and the $6N + 2$ bound, and identifies the literal fiber naturally with maps from $K[T]/(P)$ into every commutative test algebra. It also proves that the representing quotient is finite étale of rank N , proves the independent-parameter inverse polynomial irreducible of degree N , and constructs the explicit comparison

$$K(x, y, z) \simeq K(\Pi, B)(C)[S]/(E_{\Pi, B, C})$$

over the induced target-field embedding. Its finrank calculation therefore machine-checks the actual geometric degree. The formal development also checks that the determinant-one normalization is an invertible target rescaling and exposes one theorem combining determinant, geometric degree, literal fiber, naturality, finite étaleness, rank, and the degree bound. Monogenicity for arbitrary finite étale algebras, symmetric monodromy, Hilbert irreducibility, exact nonproperness and boundary-sheet accounting, the degree-two theorem, and the remaining classical arithmetic inputs are ordinary proofs. For the quintic, Lean proves the exact fiber, rank, finite étaleness, absence of rational points, and the existence of points over $\mathbb{R}$ and every $\mathbb{Q}_p$.

2 Which finite étale algebras occur?

Let K be a field of characteristic zero. A polynomial map

$$F : \mathbb{A}_K^m \longrightarrow \mathbb{A}_K^m$$

is a *Keller map* if $\det DF \in K^*$. Its geometric degree is

$$\text{gdeg}(F) = [K(x_1, \dots, x_m) : K(F_1, \dots, F_m)].$$

A Keller map is étale and hence quasi-finite. Every fiber over a K -point is therefore a finite étale K -scheme, possibly empty.

Definition 2.1 (Keller fiber). *A nonzero finite étale K -algebra A is a Keller fiber if there are a Keller map $F : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^m$ and a point $y \in \mathbb{A}^m(K)$ such that*

$$F^{-1}(y) \simeq \text{Spec } A, \quad \dim_K A = \text{gdeg}(F).$$

The equality of the two degrees is the fullness condition. It says that the special fiber contains all generic inverse sheets. Without it, a fiber may be shorter because some sheets have escaped through the nonproperness boundary.

The construction used in the proof can be read from left to right:

$$\boxed{\text{squarefree } P(T) \longmapsto G(S) = P(a + S) - P(a) \longmapsto \tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P).}$$

The last arrow is an isomorphism of schemes. It is not merely a matching of K -points or a count after passing to an algebraic closure.

Lemma 2.2 (Degree two is impossible, over every characteristic-zero field). *Let K be a field of characteristic zero, let $m \geq 1$, and let*

$$F = (F_1, \dots, F_m) \in K[X_1, \dots, X_m]^m \quad \text{satisfy} \quad \det \left(\frac{\partial F_i}{\partial X_j} \right) \in K^*.$$

Then the induced function-field extension cannot have degree two:

$$[K(X_1, \dots, X_m) : K(F_1, \dots, F_m)] \neq 2.$$

Proof. We give the scalar-extension and descent steps because the external theorem used here is most conveniently cited over $\mathbb{C}$.

The precise external input. Campbell's unnumbered theorem [3, Theorem, p. 244] states that a polynomial map $H : \mathbb{C}^m \rightarrow \mathbb{C}^m$ has a polynomial inverse if its Jacobian determinant never vanishes and the induced extension

$$\mathbb{C}(H_1, \dots, H_m) \subset \mathbb{C}(X_1, \dots, X_m)$$

is normal. In characteristic zero “normal” here is equivalent to “Galois,” because the finite function-field extension is separable. Razar and, independently, Wright later gave algebraic treatments of this Galois case [9, 13]; the proof below needs only Campbell's stated complex theorem.

Generic degree survives scalar extension. Assume for contradiction that the displayed degree over K is two. Let $K_0 \subset K$ be the subfield generated over $\mathbb{Q}$ by the finitely many coefficients of the F_i , and write F_0 for the same formulas over K_0 . The nonzero Jacobian determinant makes

$$\varphi_0 : K_0[Y_1, \dots, Y_m] \longrightarrow K_0[X_1, \dots, X_m], \quad Y_i \longmapsto F_i, \quad (2.1)$$

injective and generically finite. Indeed, the Jacobian criterion in characteristic zero makes the F_i algebraically independent, so both fraction fields in (2.1) have transcendence degree m and their extension is finite. If its degree is d_0 , there is a nonempty principal open U_0 of the target on which F_0 is finite locally free of rank d_0 : first clear denominators in the finite extension of fraction fields, and then apply generic freeness. For every field extension E/K_0 , the base change over the nonempty open $(U_0)_E$ is again finite locally free of rank d_0 . Taking generic fibers gives

$$[E(X_1, \dots, X_m) : E(F_1, \dots, F_m)] = d_0. \quad (2.2)$$

Applying (2.2) first to $E = K$ shows $d_0 = 2$.

Why degree two gives Campbell's field hypothesis. Because K_0 is finitely generated over $\mathbb{Q}$, choose an embedding $K_0 \hookrightarrow \mathbb{C}$. Equation (2.2) with $E = \mathbb{C}$ says that

$$\mathbb{C}(X_1, \dots, X_m) / \mathbb{C}(F_1, \dots, F_m)$$

has degree two. It is separable in characteristic zero. Every separable quadratic field extension is normal (the second root of the minimal polynomial of an element is its trace minus that element), hence it is Galois. Thus the complexified map satisfies exactly the hypotheses of Campbell's theorem and is a polynomial automorphism.

Faithfully flat descent of the inverse. The base change of (2.1) along $K_0 \hookrightarrow \mathbb{C}$ is therefore an isomorphism. Since $\mathbb{C}$ is faithfully flat over K_0 , tensoring the kernel and cokernel of φ_0 with $\mathbb{C}$ detects their vanishing. Hence φ_0 itself is an isomorphism. After base change to K , the original F is a polynomial automorphism, so $K(F_1, \dots, F_m) = K(X_1, \dots, X_m)$ and its function-field degree is one. This contradicts the assumed degree two. $\square$

Theorem 2.3 (Prescribed finite-étale realization). *Let $P \in K[T]$ be squarefree of degree $N \geq 3$. Choose $a \in K$ outside the finite zero set of $P'P'''$, and put*

$$G(S) = P(a + S) - P(a).$$

Let $F_G = (\Pi, B, C)$ be the quadratic-gauge map of (3.3), and define

$$\tilde{F}_G = L \circ F_G, \quad L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right).$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N, \quad \max_i \deg(\tilde{F}_G)_i \leq 6N + 2.$$

The Galois closure of the induced degree- N function-field extension has geometric and arithmetic Galois group S_N . At

$$y_{P,a} = \left(1, 0, -\frac{2P(a)}{P'(a)}\right) \quad (2.3)$$

its full fiber is

$$\boxed{\tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P)}.$$

More explicitly, for every commutative K -algebra R there is a bijection

$$\text{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \{u \in R^3 : \tilde{F}_G(u) = y_{P,a}\},$$

natural in R .

Consequently, every finite étale K -algebra of rank at least three occurs as a full fiber of a Jacobian-one map in affine three-space.

Proof. The polynomials P' and P''' are nonzero in characteristic zero when $N \geq 3$. Their product has only finitely many roots, while K is infinite, so an admissible a exists. The linear and cubic Taylor coefficients of G are $P'(a)$ and $P'''(a)/3!$, and the quadratic gauge of Section 3 is therefore defined.

At the target (2.3), its inverse polynomial is

$$G(S) - \frac{P'(a)}{2} \left(-\frac{2P(a)}{P'(a)} \right) = P(a + S).$$

Translation preserves squarefreeness, so Corollary 3.4 identifies the full fiber with $\text{Spec } K[S]/(P(a + S))$.

The translation back to the original polynomial is explicit:

$$K[S]/(P(a + S)) \longrightarrow K[T]/(P(T)), \quad \bar{S} \longmapsto \bar{T} - a,$$

with inverse $\bar{T} \mapsto \bar{S} + a$. Composing this with the natural equivalence of Corollary 3.4 gives, for every commutative K -algebra R ,

$$\text{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \tilde{F}_G^{-1}(y_{P,a})(R).$$

Thus the isomorphism includes the complete scheme structure and commutes with base change.

The linear map L fixes the chosen target because its second coordinate is zero. It multiplies the Jacobian by $-1/2$ and changes neither geometric degree nor coordinate degree. Corollary 3.10 therefore gives the determinant-one statement and the bound $6N + 2$. Since both the quotient algebra and the generic inverse problem have degree N , the fiber is full. The symmetric-monodromy assertion follows from Theorem 3.7.

For a general finite étale algebra of rank at least three, use the monogenicity lemma in Section 4 and apply the polynomial case just proved. $\square$

Theorem 2.4 (Classification of full Keller fibers). *For $N \geq 1$, let $\text{FullFib}_{3,N}(K)$ be the replete full subcategory of finite étale K -schemes whose objects are the schemes isomorphic to full fibers*

$$F^{-1}(y), \quad F : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3, \quad \text{gdeg}(F) = N, \quad y \in \mathbb{A}^3(K).$$

Then

$$\text{FullFib}_{3,N}(K) = \begin{cases} \text{FinEt}_K^N, & N = 1 \text{ or } N \geq 3, \\ \emptyset, & N = 2, \end{cases}$$

where FinEt_K^N is the full subcategory of rank- N finite étale K -schemes. In particular, for $N \geq 3$, a finite K -scheme occurs as a full fiber of a geometric-degree- N Keller map in $\mathbb{A}_K^3$ if and only if it is finite étale of rank N .

Equivalently on coordinate algebras, every rank- N finite étale K -algebra occurs for $N = 1$ or $N \geq 3$, and none occurs for $N = 2$. Consequently, the possible nonzero ranks of full Keller fibers are exactly

$$\boxed{1, 3, 4, 5, \dots}.$$

Proof. The Jacobian condition makes F étale. Hence every fiber over a K -point is a finite étale K -scheme, and the fullness equality makes its rank equal to $\text{gdeg}(F) = N$. This proves the necessary direction in every degree. The identity map realizes the unique rank-one object. Theorem 2.3 realizes every rank- N finite étale object in affine three-space for every $N \geq 3$, while Lemma 2.2 shows that no geometric-degree-two Keller map exists. This last exclusion concerns the generic function-field degree before any target point is selected. Thus an empty special fiber cannot create a rank-two object; it is not a nonzero finite étale scheme at all. Nor can a shorter special fiber of a map of some other geometric degree: such a fiber fails the defining fullness equality and is not an object of $\text{FullFib}_{3,2}(K)$. Empty and non-full fibers therefore have no bearing on the asserted rank set. $\square$

Remark 2.5 (What the categorical statement remembers). *The category $\text{FullFib}_{3,N}(K)$ is the essential image inside finite étale K -schemes: it remembers the resulting fiber scheme, not a chosen presentation by a map–target pair (F, y) . Different pairs may realize the same object. If one works with finite étale algebras rather than schemes, the corresponding statement is contravariant under Spec .*

3 The quadratic gauge

The prescribed-algebra realization theorem rests on one explicit construction. Let

$$G(S) = g_1 S + g_2 S^2 + \dots + g_N S^N \in K[S], \quad g_1 g_3 g_N \neq 0. \quad (3.1)$$

In source coordinates (x, y, z) define

$$t = 1 + xy, \quad q = t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t). \quad (3.2)$$

The associated map $F_G = (\Pi, B, C) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is

$$\begin{aligned} \Pi &= tq, \\ B &= y + 3 \frac{g_3}{g_1} xq + 2 \frac{g_2}{g_1} tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \\ C &= x(5 - 3t) - \frac{g_3}{g_1} x^3 z - \sum_{k=4}^N (k - 2) \frac{g_k}{g_1} (xq)^k. \end{aligned} \quad (3.3)$$

Although the formula is long, it is polynomial and completely mechanical in the coefficients of G .

The smallest seed recovers the counterexample that motivated this work. Take

$$G(S) = S + S^3.$$

Then

$$q = (1 + xy)^2 z + y^2(4 + 3xy),$$

and (3.3) becomes, coordinate for coordinate,

$$\begin{aligned}\Pi &= (1 + xy)^3 z + y^2(1 + xy)(4 + 3xy), \\ B &= y + 3x(1 + xy)^2 z + 3xy^2(4 + 3xy), \\ C &= 2x - 3x^2 y - x^3 z.\end{aligned}\tag{3.4}$$

This is exactly the map announced by Alpöge [1]: there is no change of source coordinates, target coordinates, or scaling hidden in the comparison. In particular, the general determinant theorem below specializes directly to its Jacobian determinant -2 .

Remark 3.1 (The normalized marked-factor origin). *As explained geometrically by Tao [12], begin with multiplication of a binary linear form and a binary quadratic form,*

$$(L, Q) \mapsto LQ, \quad \text{Sym}^1(K^2) \times \text{Sym}^2(K^2) \longrightarrow \text{Sym}^3(K^2).$$

The product is unchanged by $(L, Q) \mapsto (\lambda L, \lambda^{-1} Q)$. Since

$$\text{Res}(\lambda L, \lambda^{-1} Q) = \lambda \text{Res}(L, Q),$$

the equation $\text{Res}(L, Q) = 1$ fixes this relative scaling. For a generic cubic $L_1 L_2 L_3$, the three inverse branches are the three normalized factorizations obtained by marking L_1 , L_2 , or L_3 as the linear factor. Nonvanishing of the resultant says that the marked root is disjoint from the residual quadratic; equivalently, the coefficient–resultant multiplication map is étale along this normalized slice.

The seed $G(S) = S + S^3$ is an affine three-space chart of this normalized marked-factor map. The quadratic gauge should therefore be read as retaining the marked-root reconstruction of the cubic model while replacing its cubic inverse equation by the arbitrary polynomial G . Its roots label the inverse branches just as the three marked roots do in the seed case.

The point of the construction appears in the inverse problem. For a target (Π, B, C) put

$$G_\Pi(S) = g_1 S + \Pi(g_2 S^2 + g_3 S^3) + \sum_{k=4}^N g_k \Pi^k S^k\tag{3.5}$$

and

$$E_{\Pi, B, C}(S) = G_\Pi(S) - \frac{g_1}{2}(BS^2 + C).\tag{3.6}$$

Thus a three-variable inverse problem reduces to the single equation $E_{\Pi, B, C}(S) = 0$.

For $(\pi, b, c) \in K^* \times K^2$, write $\mathcal{F}_{\pi, b, c}$ for the fiber functor

$$\mathcal{F}_{\pi, b, c}(A) = \{(x, y, z) \in A^3 : F_G(x, y, z) = (\pi, b, c)\}$$

on commutative K -algebras.

Lemma 3.2 (Derivative units over test algebras). *Let $E \in K[S]$ be separable, let A be a commutative K -algebra, and let $s \in A$ satisfy $E(s) = 0$. Then $E'(s)$ is a unit of A .*

Proof. Choose $U, V \in K[S]$ with $UE + VE' = 1$. Evaluating at s gives $V(s)E'(s) = 1$. Since A is commutative, this is a two-sided inverse. $\square$

Theorem 3.3 (General localized quadratic-gauge fiber theorem). *The map F_G satisfies*

$$\det DF_G = -2, \quad \text{gdeg}(F_G) = N.$$

Its generic inverse equation is $E_{\Pi, B, C}(S) = 0$. Fix $(\pi, b, c) \in K^ \times K^2$, put $E = E_{\pi, b, c}$, and let $\overline{E'}$ be the class of E' in $K[S]/(E)$. Then*

$$\boxed{F_G^{-1}(\pi, b, c) \simeq \text{Spec}((K[S]/(E))[1/\overline{E'}])}. \quad (3.7)$$

Equivalently, for every commutative K -algebra A there is a natural bijection

$$\text{Hom}_{K\text{-alg}}((K[S]/(E))[1/\overline{E'}], A) \simeq \mathcal{F}_{\pi, b, c}(A). \quad (3.8)$$

Proof. We begin on the open set $t \neq 0$. Introduce the coordinates

$$S = \frac{x}{t}, \quad Q = y + xq, \quad D = 1 - SQ + \Pi S^2 = t^{-1}.$$

A direct differentiation gives

$$\det \frac{\partial(\Pi, S, Q)}{\partial(x, y, z)} = D^{-1}. \quad (3.9)$$

Next define

$$\beta(\Pi, S) = \frac{G'_{\Pi}(S)/g_1 - 1 - \Pi S^2}{S}. \quad (3.10)$$

The numerator is divisible by S . Substituting the definitions into (3.3) gives the two identities

$$B = Q + \beta(\Pi, S), \quad C = \frac{2G_{\Pi}(S)}{g_1} - BS^2. \quad (3.11)$$

The second identity is precisely $E_{\Pi, B, C}(S) = 0$. Differentiating it gives

$$D = \frac{G'_{\Pi}(S)}{g_1} - BS = \frac{E'_{\Pi, B, C}(S)}{g_1}. \quad (3.12)$$

At fixed Π , the Jacobian of (3.11) with respect to (S, Q) is

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2D. \quad (3.13)$$

The factors D^{-1} and $-2D$ cancel. Hence $\det DF_G = -2$ on the dense open set $t \neq 0$, and therefore everywhere.

We now identify the entire fiber. Let A be an arbitrary commutative K -algebra. An A -point (x, y, z) above (π, b, c) satisfies

$$tq = \pi.$$

Because π is a unit, both t and q are units. The chart coordinates

$$S = xt^{-1}, \quad Q = y + xq, \quad D = t^{-1}$$

are therefore defined on every point of the fiber. Equations (3.11) and (3.12) give

$$Q + \beta(\pi, S) = b, \quad E(S) = 0, \quad D = \frac{E'(S)}{g_1}.$$

Thus every source point gives a root S of E for which $E'(S)$ is a unit.

Conversely, suppose $s \in A$ satisfies $E(s) = 0$ and $E'(s) \in A^*$. Put

$$d = \frac{E'(s)}{g_1},$$

and define

$$Q = b - \beta(\pi, s), \quad t = d^{-1}, \quad x = sd^{-1}, \quad y = Q - \pi s,$$

followed by

$$q = \pi d, \quad z = d^2 \left(q - \frac{g_1}{g_3} y^2 (1 + 3t) \right).$$

The derivative identity says exactly that

$$d = 1 - sQ + \pi s^2.$$

A calculation in the ring A now gives

$$1 + xy = t, \quad t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t) = q, \quad tq = \pi.$$

The first identity in (3.11) gives $B = b$, and the root equation gives $C = c$. We have reconstructed a point of the original fiber.

The two procedures undo one another. Starting with a source point, one recovers

$$D = t^{-1}, \quad x = S/D, \quad y = Q - \pi S, \quad q = \pi D,$$

and then z from the definition of q . Starting with a root, the marked coordinate is again s . All formulas use ring operations and the inverse of the explicitly specified unit $E'(s)$, so they commute with homomorphisms of test algebras.

The universal property of $K[S]/(E)$ identifies its maps to A with roots of E in A . Localizing at $\overline{E'}$ imposes exactly the additional condition that $E'(s)$ be a unit. This proves (3.8), and Yoneda's lemma gives (3.7). Notice that this general localized statement requires no separability hypothesis: it removes precisely the ramified part of the inverse equation.

It remains to check the generic degree. Put

$$L = K(\Pi, B), \quad \lambda = \frac{g_1}{2},$$

and write

$$E_{\Pi, B, C}(S) = H_{\Pi, B}(S) - \lambda C \in L[C][S].$$

This polynomial has degree N in S . Its leading coefficient is $g_3 \Pi$ when $N = 3$ and $g_N \Pi^N$ when $N \geq 4$, so it is a unit of $L[C]$. Hence $E_{\Pi, B, C}$ is primitive over the UFD $L[C]$.

Assume it factors nontrivially over $L(C)[S]$. Gauss's lemma then gives a factorization in $L[C][S]$. Since the polynomial is linear in C , one factor, say $A(S)$, is independent of C . Comparing the coefficient of C shows that $A(S)$ divides the nonzero scalar $-\lambda$ in $L[S]$. Thus A is a unit, a contradiction. Therefore $E_{\Pi, B, C}$ is irreducible of degree N over $K(\Pi, B, C)$.

We finish by comparing that root extension with the actual source field. On the dense chart $t \neq 0$, the change of variables is birational:

$$K(x, y, z) = K(\Pi, S, Q).$$

The forward inclusion follows from the definitions. For the reverse inclusion put $D = 1 - SQ + \Pi S^2$. Then

$$t = D^{-1}, \quad x = \frac{S}{D}, \quad y = Q - \Pi S, \quad q = \Pi D, \quad (3.14)$$

and the defining equation for q recovers z rationally. Since

$$B = Q + \beta(\Pi, S),$$

we have $K(\Pi, S, Q) = K(\Pi, B, S)$. The second marked-line identity gives

$$C = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Because $K(\Pi, B, S) = K(x, y, z)$ has transcendence degree three and is generated by the three elements Π, B, S , those elements are algebraically independent. In particular, over $L_0 = K(\Pi, B)$, the element S is transcendental and the displayed expression for C is a nonconstant degree- N polynomial in S . Hence C is transcendental over L_0 : otherwise S , which is algebraic over $L_0(C)$, would be algebraic over L_0 . Thus Π, B, C are algebraically independent and $T = K(\Pi, B, C)$ is the actual target function field.

The evaluation homomorphism from the abstract target rational function field to the source field is therefore injective. Under it, $E_{\Pi, B, C}$ has the root S , and the reconstruction above gives

$$K(x, y, z) = K(\Pi, B, S) = T(S).$$

The irreducibility just proved makes $E_{\Pi, B, C}$ the minimal polynomial of S over T . Therefore

$$[K(x, y, z) : K(\Pi, B, C)] = [T(S) : T] = N,$$

which is exactly $\text{gdeg}(F_G) = N$. □

Corollary 3.4 (Finite-étale (squarefree) quadratic-gauge fibers). *In the setting of Theorem 3.3, suppose that $E = E_{\pi, b, c}$ is squarefree of degree N . Then the localization disappears and, for every commutative K -algebra A , there is a natural bijection*

$$\text{Hom}_{K\text{-alg}}(K[S]/(E), A) \simeq \mathcal{F}_{\pi, b, c}(A). \quad (3.15)$$

Consequently,

$$F_G^{-1}(\pi, b, c) \simeq \text{Spec } K[S]/(E),$$

and this is a full finite-étale Keller fiber of rank N .

Proof. Since E is squarefree, choose $U, V \in K[S]$ with $UE + VE' = 1$. Modulo E , the class of V is an inverse for $\overline{E'}$, so the localization in Theorem 3.3 is just $K[S]/(E)$. The natural equivalence and scheme isomorphism follow from that theorem. Its geometric-degree statement makes the rank- N fiber full. □

Lemma 3.5 (Morse slices of the quadratic pencil). *Let k be an algebraically closed field of characteristic zero, and let*

$$H(S) = g_1 S + \cdots + h_N S^N, \quad g_1 h_N \neq 0.$$

Then

$$f_b(S) = H(S) - \frac{g_1 b}{2} S^2$$

is Morse for all but finitely many $b \in k$.

Proof. A repeated root r of

$$H(S) - \frac{g_1}{2}(BS^2 + C)$$

cannot be zero, because $H'(0) = g_1$. Solving $E(r) = E'(r) = 0$ gives

$$B(r) = \frac{H'(r)}{g_1 r}, \quad C(r) = \frac{2H(r) - rH'(r)}{g_1}, \quad r \in \mathbb{G}_m. \quad (3.16)$$

Their differentials satisfy

$$dC = -r^2 dB. \quad (3.17)$$

Let $M = k(B(r), C(r)) \subset k(r)$. Equation (3.17), with B nonconstant, shows that $r^2 \in M$. Since $[k(r) : k(r^2)] = 2$, either $M = k(r^2)$ or $M = k(r)$. The first possibility is excluded by the pole of order one of $B(r)$ at $r = 0$: every element of $k(r^2)$ has even valuation there. Thus $M = k(r)$, so (3.16) is birational onto its image.

The rational map $r \mapsto B(r)$ has degree $N - 1$. Its only poles are the pole of order one at 0 and the pole of order $N - 2$ at infinity; the latter is present because $h_N \neq 0$. Outside finitely many values of b , the fiber $B(r) = b$ therefore consists of $N - 1$ distinct points. They are precisely the critical points of f_b .

At such a point,

$$f_b(r) = H(r) - \frac{rH'(r)}{2} = \frac{g_1}{2} C(r).$$

The normalization map in (3.16) is an isomorphism away from finitely many points of the image curve. After excluding the finitely many B -coordinates of its exceptional points, no two points in a fiber of B have the same C -coordinate. Thus the critical values of f_b are pairwise distinct, and f_b is Morse. $\square$

Lemma 3.6 (A Morse slice detects generic monodromy). *Let k be algebraically closed of characteristic zero. Fix $p \in k^*$ and $b \in k$. If $E_{p,b,C}(S)$ is separable over $k(C)$, then its splitting-field Galois group embeds, in its natural permutation representation, into the Galois group of $E_{\Pi,B,C}(S)$ over $k(\Pi, B, C)$.*

Proof. Normalize a target open set in the generic splitting field. The generic point of the line $\Pi = p$, $B = b$ belongs to that open after inverting the specialized discriminant, which is nonzero by separability. A point of the normalization above this generic point has a decomposition group D contained in the generic Galois group. The discriminant is invertible there, so the inertia group is trivial. The residue field is generated by the distinct specialized roots and is therefore the splitting field of $E_{p,b,C}$. Thus D is isomorphic to the specialized splitting-field group. After labeling the roots, that group is a subgroup of the generic group. $\square$

Theorem 3.7 (Full symmetric generic monodromy). *For every seed (3.1), not merely for a generic choice of its coefficients, the geometric and arithmetic monodromy groups of F_G are*

$$\boxed{\text{Mon}_{\text{geom}}(F_G) = \text{Mon}_{\text{arith}}(F_G) = S_N.}$$

Equivalently, the splitting field of $E_{\Pi,B,C}(S)$ over $K(\Pi, B, C)$ has Galois group S_N , and the same is true after replacing K by an algebraic closure. The conclusion is unchanged for the Jacobian-one normalization $\tilde{F}_G$.

Proof. We work first over $\overline{K}$. Fix $p \in \overline{K}^*$ and put $H(S) = G_p(S)$. Consider

$$f_b(S) = H(S) - \frac{g_1 b}{2} S^2.$$

Lemma 3.5 shows that f_b is Morse for all but finitely many $b \in \overline{K}$.

Choose one such b . The classical Morse-polynomial theorem [11, Theorem 4.4.5] gives

$$\text{Gal}(f_b(S) - T / \overline{K}(T)) = S_N.$$

Indeed, the polynomial cover is connected, its finite branch inertia groups are transpositions, and those inertia groups generate; a transitive permutation group generated by transpositions is the full symmetric group. After the substitution $T = g_1 C/2$, this is the inverse cover on the line $\Pi = p$, $B = b$. Lemma 3.6 embeds its monodromy into the generic geometric monodromy. Hence

$$S_N \leq \text{Mon}_{\text{geom}}(F_G) \leq S_N,$$

which proves equality. Finally,

$$\text{Mon}_{\text{geom}}(F_G) \triangleleft \text{Mon}_{\text{arith}}(F_G) \leq S_N$$

forces the arithmetic group to be S_N as well. A linear automorphism of the target does not change either monodromy group, so the same conclusion holds for $\tilde{F}_G$. $\square$

Corollary 3.8 (Hilbertian specializations). *Suppose that K is Hilbertian. For every seed (3.1), there are infinitely many targets in $K^* \times K^2$ whose fibers are connected full finite-étale schemes of rank N and whose splitting fields have Galois group S_N over K .*

Proof. Choose $p \in K^*$. Lemma 3.5 excludes only finitely many values of b in an algebraic closure. A Hilbertian field is infinite, so some $b \in K$ makes f_b Morse. Let L be the splitting field of $f_b(S) - T$ over $K(T)$. Serre's Morse-polynomial theorem gives

$$\text{Gal}(L/K(T)) = S_N$$

[11, Theorem 4.4.5]. Applied after extending the constants to $\overline{K}$, the same theorem gives geometric group S_N . Thus the arithmetic and geometric groups agree, their constant-field quotient is trivial, and $L/K(T)$ is regular over K .

Hilbert irreducibility in its Galois form [11, Chapter 3] now supplies infinitely many $c \in K$ for which $E_{p,b,c}$ remains irreducible and has Galois group S_N . Such a polynomial is squarefree of degree N , so Corollary 3.4 identifies it with a connected full finite-étale fiber. $\square$

Remark 3.9 (Which hypotheses are used where). *The general localized quadratic-gauge fiber theorem, Theorem 3.3, uses only that π , g_1 , and g_3 are nonzero for the fiber identification, together with the identity*

$$E'(S) = g_1(1 - bS + S\beta(\pi, S) + \pi S^2).$$

Squarefreeness is needed only to conclude that E' is invertible at every root and hence that no localization remains. The condition $g_3 \neq 0$ is what makes the source coefficient g_1/g_3 meaningful.

For a translated polynomial $P(a + S) - P(a)$, the cubic coefficient is the third Hasse derivative at a , equal to $P'''(a)/3!$ in characteristic zero. The Lean development follows the same separation.

The transposition inertia in Theorem 3.7 does not contradict the fact that F_G is étale. A repeated inverse root has $E' = 0$, so the reconstruction theorem removes it from the affine source; the corresponding valuation is centered on the nonproperness boundary. Monodromy is taken on the function-field cover, or equivalently on the finite étale restriction over a sufficiently small target open set.

Corollary 3.10 (Jacobian-one effective form). *Let*

$$L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right), \quad \tilde{F}_G = L \circ F_G.$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N.$$

For every $(\pi, b, c) \in K^* \times K^2$,

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec} \left((K[S]/(E_{\pi, b, c}))[1/\overline{E'_{\pi, b, c}}] \right).$$

If $E_{\pi, b, c}$ is squarefree of degree N , this reduces to

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec } K[S]/(E_{\pi, b, c}).$$

In particular, a target with $b = 0$ is unchanged by the normalization. Moreover,

$$\deg \Pi \leq 7, \quad \deg B \leq 6N + 2, \quad \deg C \leq 6N,$$

so

$$\boxed{\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.}$$

Proof. The linear map L has determinant $-1/2$, so the determinant, geometric degree, and localized-fiber statements follow from Theorem 3.3; the squarefree simplification follows from Corollary 3.4.

For the degree estimate, $\deg t = 2$ and $\deg q \leq 5$, hence $\deg \Pi \leq 7$. A summand of B indexed by $k \geq 4$ has degree at most

$$2 \deg t + (k - 2) + k \deg q \leq 4 + (k - 2) + 5k = 6k + 2.$$

The preceding terms have degrees at most 1, 6, and 7, so $\deg B \leq 6N + 2$. Similarly, $(xq)^k$ has degree at most $6k$, while the remaining terms of C have degree at most 4. Thus $\deg C \leq 6N$. A linear output normalization does not change these bounds. $\square$

3.1 Checking the formula coefficient by coefficient

The compact formulas in the general localized quadratic-gauge fiber theorem, Theorem 3.3, hide a simple term-by-term calculation. We include it here because this is where the map for an arbitrary degree- N polynomial is assembled. The argument treats one generic coefficient g_k at a time; it is not inferred from low-degree examples.

From (3.5) and (3.10),

$$\beta(\Pi, S) = 2\frac{g_2}{g_1}\Pi + \left(3\frac{g_3}{g_1} - 1\right)\Pi S + \sum_{k=4}^N k\frac{g_k}{g_1}\Pi^k S^{k-2}. \quad (3.18)$$

On the source chart,

$$S = \frac{x}{t}, \quad \Pi = tq, \quad Q = y + xq,$$

and therefore, for every $k \geq 4$,

$$\Pi^k S^{k-2} = t^2 x^{k-2} q^k. \quad (3.19)$$

Substitution now gives

$$\begin{aligned} Q + \beta(\Pi, S) &= y + xq + 2\frac{g_2}{g_1}tq + \left(3\frac{g_3}{g_1} - 1\right)xq + \sum_{k=4}^N k\frac{g_k}{g_1}t^2 x^{k-2} q^k \\ &= y + 3\frac{g_3}{g_1}xq + 2\frac{g_2}{g_1}tq + \sum_{k=4}^N k\frac{g_k}{g_1}t^2 x^{k-2} q^k, \end{aligned}$$

which is the displayed coordinate B .

For the third coordinate, write

$$C_* = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Using $B = Q + \beta(\Pi, S)$ and the definition of β , we obtain

$$C_* = 2S - QS^2 + \left(1 - \frac{g_3}{g_1}\right)\Pi S^3 - \sum_{k=4}^N (k-2)\frac{g_k}{g_1}\Pi^k S^k. \quad (3.20)$$

The terms with $k \geq 4$ already have the desired source form, since

$$\Pi^k S^k = (xq)^k. \quad (3.21)$$

It remains to simplify the cubic part. Put $r = g_3/g_1$. Then

$$\begin{aligned} 2S - QS^2 + (1-r)\Pi S^3 &= \frac{2xt - x^2y - rx^3q}{t^2} \\ &= \frac{x(t+1) - rx^3q}{t^2}, \end{aligned}$$

where the second line uses $xy = t - 1$. Since

$$q = t^2z + \frac{1}{r}y^2(1+3t),$$

this becomes

$$-rx^3z + \frac{x(t+1 - (t-1)^2(1+3t))}{t^2}.$$

The elementary identity

$$t+1 - (t-1)^2(1+3t) = t^2(5-3t) \quad (3.22)$$

therefore gives

$$2S - QS^2 + (1 - r)\Pi S^3 = x(5 - 3t) - \frac{g_3}{g_1}x^3z.$$

Together with (3.20) and (3.21), this is exactly the displayed coordinate C .

We derived the identities after inverting t , but both sides are polynomials in $K[x, y, z]$. Localization at the nonzero polynomial t is injective, so the identities hold everywhere. At fixed Π , their differential has determinant

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2 \left(\frac{G'_\Pi(S)}{g_1} - BS \right) = -2D.$$

The source-chart determinant is D^{-1} , and the chain rule gives $\det DF_G = -2$. This is the complete cancellation: each coefficient g_k enters independently, so the same calculation works in every degree.

3.2 The exact reduced nonproperness locus

We now make the fullness condition intrinsic. Work geometrically over $\overline{K}$. For a polynomial map $F : \mathbb{A}_{\overline{K}}^n \rightarrow \mathbb{A}_{\overline{K}}^n$, let S_F be the set of target points at which F is not finite over any Zariski neighborhood. Since source and target are affine, this is the same as failure of properness over every neighborhood. Equivalently, compactify the graph in $\mathbb{P}_{\overline{K}}^n \times \mathbb{A}_{\overline{K}}^n$; then S_F is the image of its boundary. We give S_F its reduced closed-subscheme structure. Over $\mathbb{C}$ this agrees with the usual analytic Jelonek set, defined by sequences escaping to infinity.

For a polynomial h , write $V_{\text{red}}(h)$ for the reduced closed subscheme $V(\sqrt{(h)})$. Put

$$\Delta_G(\Pi, B, C) = \text{Disc}_S(E_{\Pi, B, C}(S)). \quad (3.23)$$

Here the discriminant is the universal degree- N polynomial expression in the coefficients of $E_{\Pi, B, C}$. We do not replace it on a special slice by the discriminant of the lower-degree specialization.

Lemma 3.11 (Fiber cardinality and nonproperness in characteristic zero). *Let k be an algebraically closed field of characteristic zero and let $F : \mathbb{A}_k^n \rightarrow \mathbb{A}_k^n$ be a dominant étale polynomial map of geometric degree d . Then*

$$y \notin S_F \iff \#F^{-1}(y) = d. \quad (3.24)$$

Proof. Jelonek proves this multiplicity criterion for complex polynomial maps [5, Proposition 6]. We record why that stated complex result is sufficient here. Descend the finitely many coefficients of F and the coordinates of y to a finitely generated field $k_0/\mathbb{Q}$, and choose an embedding $k_0 \hookrightarrow \mathbb{C}$. The graph-boundary description above is compatible with extension of algebraically closed fields: a boundary fiber is empty before base change exactly when it is empty after base change. Thus membership in S_F is unchanged. Because F is étale, its fiber at y is a finite étale k_0 -algebra; its dimension, hence its number of geometric points after either algebraic closure, is also unchanged. Finally, geometric degree is invariant under this scalar extension. Proposition 6 over $\mathbb{C}$, where every local multiplicity is one, therefore gives (3.24) over k . $\square$

Theorem 3.12 (Exact reduced quadratic-gauge nonproperness). *For every seed (3.1),*

$$\boxed{S_{F_G} = V_{\text{red}}(\Delta_G)}. \quad (3.25)$$

Equivalently, their sets of geometric points satisfy $|S_{F_G}| = V_{\overline{K}}(\Delta_G)$. More precisely, the following statements hold.

1. If $\pi \neq 0$ and the multiplicity partition of $E_{\pi,b,c}$ is $(m_1, \dots, m_s)$, then

$$\#F_G^{-1}(\pi, b, c) = \#\{i : m_i = 1\}, \quad N - \#F_G^{-1}(\pi, b, c) = \sum_{m_i \geq 2} m_i. \quad (3.26)$$

2. On $\Pi = 0$, the complete geometric fiber table is

<i>target stratum</i>	$\#F_G^{-1}(0, B, C)$	<i>lost sheets</i>
$B \neq 0, BC \neq 1$	3	$N - 3$
$B \neq 0, BC = 1$	1	$N - 1$
$B = 0$	1	$N - 1$.

(3.27)

3. For $N = 3$, the raw resultant has an extraneous leading-coefficient factor:

$$\text{Res}_S(E, E') = \text{unit} \cdot \Pi \Delta_G, \quad \Delta_G(0, B, C) = \frac{g_1^4}{4} B^2 (1 - BC).$$

Hence the entire plane $\Pi = 0$ is not a nonproperness component in degree three.

4. For $N \geq 4$, put

$$e_N = N^2 - 3N - 2.$$

Then

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2 (1 - BC), \quad \kappa_G \in K^*. \quad (3.28)$$

In particular,

$$S_{F_G} = V_{\text{red}}(\Pi D_G),$$

whose underlying set is $V(\Pi) \cup V(D_G)$, and the resultant and discriminant define the same reduced closed subscheme.

Proof. Because F_G is étale, every geometric fiber is finite and reduced. Its generic cardinality is N , so Lemma 3.11 applies with $d = N$.

Suppose first that $\pi \neq 0$. Theorem 3.3 identifies the fiber with

$$\text{Spec}((\overline{K}[S]/(E_{\pi,b,c}))[1/\overline{E'_{\pi,b,c}}]).$$

Thus a distinct root contributes one affine point exactly when it is simple. This proves (3.26); it also shows that the fiber has cardinality N exactly when $\Delta_G(\pi, b, c) \neq 0$.

It remains to analyze $\Pi = 0$, where the inverse chart used above is not available. In the source, $\Pi = tq$. On $q = 0$, one has

$$B = y, \quad C = 2S - BS^2, \quad S = \frac{x}{t}.$$

The residual inverse equation is therefore

$$BS^2 - 2S + C = 0. \quad (3.29)$$

A root reconstructs on $q = 0$ precisely when $1 - BS \neq 0$. Hence this divisor contributes two points when $B \neq 0$ and $BC \neq 1$, no point when $B \neq 0$ and $BC = 1$, and one point when $B = 0$.

On $t = 0$, the relation $1 + xy = 0$ gives $y = -1/x$, while

$$q = \frac{g_1}{g_3} y^2, \quad B = -2y = \frac{2}{x}.$$

Thus $t = 0$ contributes one point when $B \neq 0$ and none when $B = 0$. Once x and y are fixed, the equation for C is linear in z with nonzero coefficient, so the point is unique. The divisors $t = 0$ and $q = 0$ do not meet. This proves (3.27).

For $N = 3$, specializing the degree-three discriminant at $\Pi = 0$ gives

$$\Delta_G(0, B, C) = \frac{g_1^4}{4} B^2 (1 - BC).$$

Its zero set is exactly the two short-fiber strata in (3.27). Since the leading coefficient is $g_3 \Pi$, the resultant has one additional factor Π .

Now let $N \geq 4$, put $r = N - 3$, and set $R = \overline{K}[B, C]$. Over $\text{Frac}(R)[[\Pi]]$, the lower Newton polygon of E has vertices

$$(0, 0), \quad (2, 0), \quad (3, 1), \quad (N, N).$$

Consequently its root valuations, with multiplicities, are

$$0 \ (2), \quad -1 \ (1), \quad -\frac{r+2}{r} \ (r).$$

The first two blocks are exactly the affine $q = 0$ and $t = 0$ branches found above; the final block consists of the $N - 3$ missing sheets. The residual equations of the three blocks are, respectively,

$$BZ^2 - 2Z + C = 0, \quad Z = \frac{g_1 B}{2g_3}, \quad g_3 + g_N Z^r = 0.$$

They are separable over $\text{Frac}(R)$: the first has discriminant $4(1 - BC)$, the second is linear, and the last is separable because the characteristic is zero and $g_3 g_N \neq 0$.

Using

$$\text{Disc}(E) = \text{lc}(E)^{2N-2} \prod_{i < j} (s_i - s_j)^2$$

and the three displayed root valuations gives

$$\text{ord}_\Pi(\Delta_G) = 2N(N - 1) - 4 - 4(r + 2) - 2(r + 2) - (r - 1)(r + 2) = N^2 - 3N - 2.$$

Separability makes this order exact. To retain the dependence on B, C , let α_1, α_2 be the roots of the first residual equation, put $b_0 = g_1 B / (2g_3)$, and let $v_1, \dots, v_r$ be the roots of $g_3 + g_N Z^r$. The leading coefficient in the root-product formula is

$$\begin{aligned} & \pm g_N^{2N-2} (\alpha_1 - \alpha_2)^2 b_0^4 \left(\prod_j v_j \right)^6 \prod_{i < j} (v_i - v_j)^2 \\ &= \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} B^2 (1 - BC). \end{aligned}$$

Indeed,

$$(\alpha_1 - \alpha_2)^2 = \frac{4(1 - BC)}{B^2}, \quad \left(\prod_j v_j \right)^6 = \left(\frac{g_3}{g_N} \right)^6, \quad \prod_{i < j} (v_i - v_j)^2 = \pm r^r \left(\frac{g_3}{g_N} \right)^{r-1}.$$

These calculations were made in $\text{Frac}(R)$, but Π -adic order in the domain $R[[\Pi]]$ is unchanged on passing to its fraction field. Hence they prove the global polynomial identity

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2 (1 - BC), \quad \kappa_G = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1} \neq 0.$$

Thus e_N is the exact global Π -adic order. After specializing (B, C) , the pointwise order is e_N when $B(1 - BC) \neq 0$ and may jump on the two displayed special strata.

No general assertion about discriminants under a drop of leading coefficient is being used. For $N = 3$, the explicit specialized formula cuts out exactly the two short-fiber strata. For $N \geq 4$, Δ_G vanishes on the whole slice $\Pi = 0$, and every fiber on that slice has fewer than N points by (3.27). On $\Pi \neq 0$, the ordinary degree- N discriminant detects exactly the repeated-root, hence short-fiber, locus. The fiber table and (3.24) now show that the zero set of Δ_G is exactly the underlying nonproperness set, and taking radicals gives (3.25). $\square$

Proposition 3.13 (Boundary normalization and explicit escape arcs). *On $\Pi \neq 0$, the reduced discriminant divisor is birationally parametrized by*

$$B = \frac{G'_\Pi(s)}{g_1 s}, \quad C = \frac{2G_\Pi(s) - sG'_\Pi(s)}{g_1}, \quad (\Pi, s) \in \mathbb{G}_m^2. \quad (3.30)$$

If s_0 is a root of multiplicity $m \geq 2$ over (π, b, c) , every one of the m lost sheets is represented, after a finite base change, by a Laurent-series source arc. With root displacement as uniformizer, its generic valuations are

$$\text{ord}(x) = -(m - 1), \quad \text{ord}(q) = m - 1, \quad \text{ord}(z) \geq m - 1.$$

If $N \geq 4$, put $d = N - 3$ and $h = \gcd(d, 2)$. At the generic point of $\Pi = 0$, the d lost sheets form h boundary primes, each with ramification index d/h . After the common base change $\Pi = u^d$, their valuations are

$$\begin{array}{c|cccccccc} & S & D & t & x & y & q & z \\ \hline \text{ord}_u & -(N - 1) & -(N + 1) & N + 1 & 2 & -2 & -4 & -(2N + 6). \end{array} \quad (3.31)$$

For $N = 3$ this generic boundary block is absent. On the special stratum $B = 0$, two further missing sheets have

$$\Pi = u^2, \quad S \sim au^{-1}, \quad a^2 = -\frac{g_1}{g_3}, \quad (\text{ord}_u x, \text{ord}_u y, \text{ord}_u q, \text{ord}_u z) = (-1, 1, 2, 2). \quad (3.32)$$

Together with the multiplicity-block arcs above, these account for every lost sheet on every stratum of (3.27).

Proof. Equations (3.30) are $E = E' = 0$ solved for B, C . Their birationality was proved in the monodromy argument around (3.16).

For a repeated root, set $s(u) = s_0 + u$ and keep π, b fixed while putting

$$c(u) = \frac{2G_\pi(s(u))}{g_1} - b s(u)^2, \quad d(u) = \frac{G'_\pi(s(u))}{g_1} - b s(u).$$

Then $E_{\pi, b, c(u)}(s(u)) = 0$ identically and $\text{ord}_u d = m - 1$. The exact reconstruction formulas

$$x = \frac{s}{d}, \quad y = b - \beta(\pi, s) - \pi s, \quad q = \pi d, \quad z = \pi d^3 - \frac{g_1}{g_3} y^2 (d^2 + 3d)$$

give the stated valuations. Applying Newton–Puiseux to the other roots in the same multiplicity block gives all m arcs after a finite base change.

Suppose $N \geq 4$. For $\Pi = 0$, the final Newton edge has horizontal length d and endpoint equation

$$g_3 + g_N Z^d = 0.$$

Equivalently its binomial model is

$$W^d = \text{unit} \cdot \Pi^{d+2}.$$

It has $h = \gcd(d, 2)$ primes, each of ramification index d/h . Normalizing by

$$v(\Pi) = \frac{d}{h}, \quad v(S) = -\frac{d+2}{h}$$

gives

$$v(D) = -\frac{d+4}{h}, \quad v(x) = \frac{2}{h}, \quad v(y) = -\frac{2}{h}, \quad v(q) = -\frac{4}{h}, \quad v(z) = -\frac{2d+12}{h}.$$

Multiplication by h , corresponding to the common base change $\Pi = u^d$, yields (3.31).

If $B = 0$, the first Newton edge changes from $(0, 0) - (2, 0)$ to $(1, 0) - (3, 1)$. Its two nonzero roots satisfy

$$\Pi = u^2, \quad S \sim au^{-1}, \quad g_1 + g_3 a^2 = 0.$$

Here D and $t = D^{-1}$ have order zero. Substitution in the exact reconstruction formulas gives (3.32). When $BC = 1$ with $B \neq 0$, the two additional missing sheets instead form the finite double-root block already covered by the first part of the proposition. $\square$

Corollary 3.14 (Intrinsic fullness of the distinguished fiber). *For the seed $G(S) = P(a + S) - P(a)$ used in Theorem 2.3, the distinguished target $y_{P,a}$ lies in the maximal open subset on which $\tilde{F}_G$ is finite étale of degree N .*

Proof. At $y_{P,a}$, the inverse polynomial is $P(a + S)$. Hence

$$\Delta_G(y_{P,a}) = \text{Disc}(P(a + S)) = \text{Disc}(P) \neq 0.$$

Theorem 3.12 places the target outside the nonproperness hypersurface. Proper and étale is finite étale, and the degree is N . $\square$

Remark 3.15 (Polynomial curves). *The parametrization (3.30), with s fixed, gives polynomial curves in Π covering the discriminant component, while the component $\Pi = 0$ is covered by affine lines. Thus the explicit geometry realizes directly the general theorem that a complex Jelonek set is a uniruled hypersurface covered by polynomial curves of bounded degree [5, 6].*

4 From a polynomial to an arbitrary finite étale algebra

The main theorem was stated for a squarefree polynomial. To reach an arbitrary finite étale algebra, we only need the following standard form of the primitive-element argument. It works for products of fields as well as for a single field.

Lemma 4.1 (Monogenicity). *Every finite étale algebra over an infinite field is generated by one element. Equivalently, if A has rank N , then*

$$A \simeq K[T]/(P)$$

for a squarefree polynomial $P \in K[T]$ of degree N .

Proof. For $\theta \in A$, let $\chi_\theta(T)$ be the characteristic polynomial of multiplication by θ on the K -vector space A , and set

$$\Delta(\theta) = \text{disc}(\chi_\theta).$$

This is a polynomial function on the affine space underlying A . After extending scalars to a separable closure, identify $A \otimes_K K^{\text{sep}}$ with $(K^{\text{sep}})^N$. In these coordinates,

$$\Delta(\theta_1, \dots, \theta_N) = \prod_{i < j} (\theta_i - \theta_j)^2,$$

so Δ is not the zero polynomial. Since K is infinite, some $\theta \in A$ satisfies $\Delta(\theta) \neq 0$. Its characteristic polynomial has N distinct geometric roots, and therefore its minimal polynomial has degree N . Hence $K[\theta] = A$, with squarefree minimal polynomial P . $\square$

Proposition 4.2 (Compatibility with extension of scalars). *Let $K \hookrightarrow K'$ be an extension of characteristic-zero fields. For a squarefree $P \in K[T]$ and an admissible translation parameter $a \in K$, the map $\tilde{F}_G$ of Theorem 2.3, after coefficientwise extension to K' , is exactly the map obtained from the same P and a over K' . Its distinguished fiber satisfies*

$$\tilde{F}_{G,K'}^{-1}(y_{P,a}) \simeq \text{Spec}((K[T]/(P)) \otimes_K K') \simeq \text{Spec } K'[T]/(P).$$

Proof. Every formula in (3.2)–(3.3), the output normalization L , and the target (2.3) is obtained from the coefficients of P and the element a by field operations. The construction therefore commutes with scalar extension. The displayed fiber identity is the usual base-change isomorphism for a polynomial quotient, combined with the prescribed-algebra realization theorem, Theorem 2.3. $\square$

Corollary 4.3 (Number fields as connected Keller fibers). *Every finite separable field extension L/K of degree at least three occurs as a connected full fiber of an explicit Jacobian-one polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ of coordinate degree at most $6[L : K] + 2$.*

Proof. Choose a primitive element of L/K and apply Theorem 2.3 to its separable minimal polynomial. $\square$

Corollary 4.4 (Arithmetic data are preserved). *Let A be a finite étale algebra of rank at least three over a number field. The realization supplied by Theorem 2.3 preserves the algebra A itself. In particular it preserves:*

- *connectedness and field decomposition;*
- *real signature;*
- *splitting field and Galois action on geometric points;*
- *the local algebras $A \otimes_K K_v$ at every place;*
- *for any fixed integral model, factorization types away from its bad primes;*
- *the existence or nonexistence of rational and local points.*

Thus an arithmetic example built as a finite étale algebra of rank at least three can be used without alteration as a Keller fiber. This works because the fiber is $\text{Spec } A$ itself, not merely a set with the same number of points. Proposition 4.2 gives the corresponding compatibility with local fields and other scalar extensions.

5 The smallest Hasse-principle failure

For a number field K , call a Keller fiber A a *Hasse failure* if $\text{Spec } A$ has a point over every completion of K but no K -rational point. In this section d_{HP} denotes the least rank of such a fiber over $\mathbb{Q}$. We show that $d_{\text{HP}} = 5$.

Lemma 5.1 (Degree-four barrier). *A nonempty finite étale $\mathbb{Q}$ -scheme of degree at most four which has a point over every $\mathbb{Q}_p$ has a rational point.*

Proof. Let X be such a scheme and let L split it. At every unramified prime, $X(\mathbb{Q}_p) \neq \emptyset$ exactly when the Frobenius element fixes a point of the finite $\text{Gal}(L/\mathbb{Q})$ -set $X(L)$.

Suppose first that X is connected of degree greater than one. The Galois action is transitive and contains a derangement. Indeed, Burnside's lemma says that the average number of fixed points is one, whereas the identity fixes more than one point. Some group element therefore fixes none. Chebotarev produces an unramified prime with that Frobenius class, contradicting local solubility.

Now suppose that X is disconnected and has no rational component. In total degree at most four, the only remaining possibility is a disjoint union of two quadratic fields. If they are equal, choose an inert prime. If they are distinct, their compositum has Galois group $C_2 \times C_2$ and contains an element nontrivial in both quadratic quotients. Chebotarev again gives a prime inert in both components. Thus every everywhere locally soluble finite étale scheme of degree at most four contains $\text{Spec } \mathbb{Q}$. $\square$

We use the classical polynomial of Berend and Bilu,

$$P_5(T) = (T^3 - 19)(T^2 + T + 1). \quad (5.1)$$

It appeared in their study of polynomials with roots modulo every integer [2].

Theorem 5.2 (Optimal Hasse-failing Keller fiber). *There is an explicit Keller map $F : \mathbb{A}_{\mathbb{Q}}^3 \rightarrow \mathbb{A}_{\mathbb{Q}}^3$ of geometric degree five and a rational target y such that*

$$F^{-1}(y)(\mathbb{Q}) = \emptyset, \quad F^{-1}(y)(\mathbb{Q}_v) \neq \emptyset \quad \text{for every place } v.$$

No Hasse-failing Keller fiber over $\mathbb{Q}$ has rank below five. Hence

$$\boxed{d_{\text{HP}} = 5}.$$

Proof. Write

$$g(T) = T^3 - 19, \quad q(T) = T^2 + T + 1.$$

The following table is the complete nonarchimedean local-solubility audit. In a generic row, the displayed residue root is simple and hence lifts by Hensel's lemma. For a finite base-case audit we remove 2, 3, 19 from the generic rows and give a direct witness at each of them.

prime p	factor	residue or approximate root	Hensel check
$p \notin \{2, 3, 19\}, p \equiv 2 \pmod{3}$	g	$a^3 = 19 \text{ in } \mathbb{F}_p$	$g'(a) = 3a^2 \neq 0$
$p \notin \{2, 3, 19\}, p \equiv 1 \pmod{3}$	q	$\zeta \in \mathbb{F}_p^\times, \text{ord}(\zeta) = 3$	$q'(\zeta) = 2\zeta + 1 \neq 0$
$p = 2$	g	$a = 1$	$g(1) = -18, g'(1) = 3 \in \mathbb{Z}_2^\times$
$p = 3$	g	$a = -2$	$v_3(g(-2)) = 3 > 2 = 2v_3(g'(-2))$
$p = 19$	q	$a = 7$	$q(7) = 57, q'(7) = 15 \in \mathbb{Z}_{19}^\times$

Here are the justifications behind the two generic rows. If $p \equiv 2 \pmod{3}$, then $\gcd(3, p-1) = 1$, so cubing is an automorphism of $\mathbb{F}_p^\times$; since $p \neq 19$, it gives a nonzero a with $a^3 = 19$, and $p \neq 3$ makes

$3a^2$ nonzero. If $p \equiv 1 \pmod{3}$, the cyclic group $\mathbb{F}_p^\times$ contains an element ζ of order three. Then $\zeta \neq 1$ and

$$0 = \zeta^3 - 1 = (\zeta - 1)q(\zeta).$$

The root is simple because the two roots of q can coincide only when its discriminant -3 vanishes, which is excluded. The rows $p = 2$ and $p = 19$ use ordinary simple-root Hensel. The $p = 3$ row uses the strong form of Hensel's lemma: $g(-2) = -27$ and $g'(-2) = 12$, so $v_3(g(-2)) > 2v_3(g'(-2))$.

This covers every finite prime, since every prime outside $\{2, 3, 19\}$ is congruent to 1 or 2 modulo 3. At the real place, $g(2) = -11 < 0 < 8 = g(3)$, so the intermediate value theorem supplies a real root.

There is no rational root: the rational-root theorem excludes a root of g (none of $\pm 1, \pm 19$ is a root), while q has discriminant -3 . Moreover, $P_5 = gq$ is squarefree: both factors are squarefree and they are coprime, since a root of q has cube 1 and therefore cannot be a root of g . The map below has geometric degree five and, by Corollary 3.4, has fiber exactly $\text{Spec } \mathbb{Q}[T]/(P_5)$. Its local and global points are consequently exactly those just audited. Finally, Lemma 5.1 applies to every complete regular fiber of rank at most four and proves rank-minimality. $\square$

For reference, here is the actual map. It comes directly from the normalized gauge, so its determinant does not require a separate guess or search. Put

$$G(T) = P_5(T) - P_5(0) = T^5 + T^4 + T^3 - 19T^2 - 19T.$$

Thus

$$(g_1, g_2, g_3, g_4, g_5) = (-19, -19, 1, 1, 1).$$

With $t = 1 + xy$ and

$$q = t^2z - 19y^2(1 + 3t),$$

Theorem 3.3 gives the gauge map

$$\begin{aligned} \Pi &= tq, \\ B_0 &= y - \frac{3}{19}xq + 2tq - \frac{4}{19}t^2x^2q^4 - \frac{5}{19}t^2x^3q^5, \\ C_0 &= x(5 - 3t) + \frac{1}{19}x^3z + \frac{2}{19}(xq)^4 + \frac{3}{19}(xq)^5, \end{aligned}$$

with $\det D(\Pi, B_0, C_0) = -2$. Apply the target linear automorphism

$$L(\Pi, B_0, C_0) = (\Pi, 19B_0, 19C_0).$$

The resulting denominator-free map $F = L \circ F_G$ is

$$\begin{aligned} F_1 &= tq, \\ F_2 &= 19y - 3xq + 38tq - 4t^2x^2q^4 - 5t^2x^3q^5, \\ F_3 &= 19x(5 - 3t) + x^3z + 2(xq)^4 + 3(xq)^5. \end{aligned} \tag{5.2}$$

Consequently,

$$\boxed{\det DF = \det(L)(-2) = 19^2(-2) = -722.}$$

The normalized target $(1, 0, -2)$ is sent to

$$y = (1, 0, -38).$$

At that target the inverse equation is

$$G(T) - \frac{-19}{2}(-2) = G(T) - 19 = P_5(T).$$

The finite-étale (squarefree) fiber corollary, Corollary 3.4, therefore gives

$$\boxed{F^{-1}(1, 0, -38) \simeq \operatorname{Spec} \mathbb{Q}[T]/((T^3 - 19)(T^2 + T + 1))}. \quad (5.3)$$

So the smallest Hasse-failing Keller fiber is not only existential; its map, target, determinant, and quotient algebra are all explicit.

6 Consequences, scope, and further directions

The main theorem is a prescribed-fiber realization result. Given a finite étale algebra, it builds a Keller map whose full fiber is that algebra. The classification by rank is a useful corollary, but it is not the main point: two finite étale algebras of the same rank can have different field decompositions, Galois actions, signatures, and local behavior, and the construction preserves all of that data.

Several consequences are immediate. A squarefree polynomial of degree at least three becomes an exact inverse polynomial after one translation; an irreducible polynomial gives a connected fiber; and a degree- N input is realized in $\mathbb{A}^3$ with coordinate degree at most $6N + 2$. Every map produced by the construction has geometric and arithmetic generic monodromy S_N . Over a Hilbertian field, each such fixed map has infinitely many connected full fibers with splitting-field group S_N .

The construction also has a simple geometric starting point. The cubic seed is an affine chart of normalized multiplication

$$(L, Q) \longmapsto LQ, \quad \operatorname{Res}(L, Q) = 1,$$

with a marked linear factor of a binary cubic. Its three generic inverse branches are the three choices of marked root, and the resultant condition is the local-invertibility mechanism. The quadratic gauge carries this marked-root reconstruction from the cubic seed to an arbitrary inverse polynomial G ; this is the geometric bridge from the announced counterexample to the all-degree realization theorem [12].

Fullness also has an intrinsic geometric interpretation in this family. Theorem 3.12 identifies the exact reduced nonproperness locus with the inverse-discriminant zero set, and Corollary 3.14 places every distinguished target in its complement. The partition formula (3.26), the complete table (3.27), and Proposition 3.13 account for the affine and boundary sheets on every displayed stratum.

This is a transfer statement, not an inverse Galois theorem. It does not say that arbitrary local specifications can always be globalized. It says that once a finite étale algebra with the desired arithmetic properties exists, the Keller construction preserves it exactly. Nor does the monodromy theorem say that one fixed seed realizes every rank- N finite étale algebra: the prescribed algebra is used to choose the seed. Its strengthening is that every resulting cover, including one built for a highly special algebra, has maximal generic monodromy.

Relation to earlier work. To my knowledge, the following prescribed-algebra statement has not appeared before: every finite étale K -algebra of rank at least three occurs as a *full scheme fiber* of an explicit Keller map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$. The rank set $\{1, 3, 4, \dots\}$ follows from this theorem together with the known degree-two obstruction. I searched for earlier results up to 24 July 2026, but this priority statement is necessarily qualified: searches can miss unpublished work or a theorem stated in different language.

The closest constructions have a different scope. Gallagher gives explicit Keller maps in every geometric degree together with complete split rational fibers [4]; these realize K^N , rather than an arbitrary finite étale K -algebra. Miranda–Neto studies radicality and maximal-ideal decompositions of fibers that already come from Keller maps [8]. Lipton and Markakis connect rational images with Hilbert irreducibility [7], and Shaska studies graded Keller maps and arithmetic thinness [10]. The monodromy argument here uses the classical theorem that a Morse polynomial has symmetric Galois group over the value-function field [11, Theorem 4.4.5].

Independent checks and formalization. The SymPy checker

`scripts/verify_universal_quadratic_gauge.py`

verifies the source chart, the marked-line identities, the Jacobian cancellation, every generic coefficient contribution, and the degree bound. The independent Singular checker expands the complete degree-six map over a rational function field in six coefficients, obtains determinant -2 , and checks the degree profile $(7, 38, 36)$. Concrete degree-three, degree-four, and degree-five regressions reconstruct the polynomial quotient in both directions. The additional checker

`scripts/verify_quadratic_gauge_nonproperness.py`

independently checks the cubic and quartic discriminant specializations, the direct $q = 0$ and $t = 0$ charts, discriminant orders and saturated-slice coefficients through degree ten, and the uniform Newton and boundary-valuation ledger through degree sixty-four.

The Lean project in

`formal/finite-etale-keller`

contains no `sorry` and introduces no project-specific axioms. It constructs the all-degree map, proves its normalized Jacobian and coordinate degree bound, identifies the literal distinguished fiber naturally over every commutative test algebra, and proves that its representing quotient has the expected rank and is finite étale. It also proves irreducibility and degree of the independent-parameter inverse equation, algebraic independence of the three map coordinates, and injectivity of the induced function-field pullback. It then constructs, over the canonical target presentation $K(\Pi, B)(C)$, the algebra equivalence

$$K(x, y, z) \simeq K(\Pi, B)(C)[S]/(E_{\Pi, B, C})$$

and transfers the inverse-root finrank to prove the actual geometric degree N . A separate theorem checks the inverse target rescaling from the Jacobian-one map back to the raw gauge. The public theorem

`automaticRealization_pageOne`

simultaneously records determinant one, geometric degree N , the natural literal-fiber representation by $K[T]/(P)$, finite étaleness, finiteness, rank N , and the $6N + 2$ bound.

The Lean axiom reports for these declarations contain only `propext`, `Classical.choice`, and `Quot.sound`. Monogenicity, symmetric monodromy, Hilbert irreducibility, the exact nonproperness and boundary-sheet accounting, the Campbell–Razar–Wright theorem, and the number-theoretic local-solubility arguments remain ordinary mathematical proofs or inputs. For the quintic example, Lean verifies the exact fiber, its rank and finite étaleness, the absence of rational points, and the existence of points over $\mathbb{R}$ and every $\mathbb{Q}_p$. The Chebotarev-based rank-minimality argument in Theorem 5.2 remains an ordinary proof.

Addendum: arithmetic left for later

The quadratic gauge has more structure than is needed for the realization theorem. The exact nonproperness boundary is proved in Subsection 3.2. The following arithmetic direction is recorded so that its exact inverse-family computation is not hidden, but it is not used as a theorem in this paper.

One map with many Hasse failures. There is also a fixed quintic gauge for which exact algebra suggests the family

$$\operatorname{Spec} \mathbb{Q}[X]/((X^3 - \ell)(X^2 + X + 1)), \quad \ell \equiv 1 \pmod{27},$$

as fibers at pairwise distinct rational targets. The inverse-polynomial identity is exact, and the expected local argument combines Hensel lifting, the residue-class dichotomy used in Theorem 5.2, and primes in an arithmetic progression. A satisfactory treatment should audit all local cases and the claimed counting asymptotic as a separate arithmetic theorem. For the present paper, the single optimal quintic fiber is enough.

Other constructions in the accompanying repository place the same finite étale algebra inside weighted and cancellation-type Keller maps that need not be stably equivalent. Their comparison belongs with those deferred questions, not with the prescribed-fiber theorem proved here.

Provenance and AI assistance. The recent three-dimensional counterexample was announced by Alpöge, who credited Akhil Mathew for suggesting the problem and Fable for producing the example [1]. Gallagher independently developed a weighted all-degree viewpoint and a public atlas of complete rational fibers [4]. The present proof uses a root-engineered quadratic gauge that contains the announced map as the seed $S + S^3$. Berend and Bilu supplied the classical interjective quintic, while Campbell, Razar, and Wright supplied the degree-two obstruction [3, 9, 13].

I used generative AI tools during exploration, symbolic-checker development, proof review, literature search, Lean formalization, and manuscript editing. They were useful as fast but fallible collaborators. I checked the stated mathematical claims, rewrote the exposition, and take responsibility for the paper.

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